

Unify Learn — Developer Report (4 October 2026)

Covers the 1 October 2026 test bug list (14 confirmed: 7 critical P0, 7 important P1), everything fixed since, what is left, and what still needs live verification — and why.

Current version: **v1.11.1**. Every change below was verified with `tsc --noEmit` both sides plus a production `vite build` before commit.

P0 fixes (critical — blocked student testing)

- **BUG-001 Contributor sees all courses (v1.8.1)**. Server gates (`requireCourseAccess` , 403 for collaborators outside their teaching enrollments) on weeks, topics, notes and publish; Browse/Studio only list assigned courses; admin can assign courses per user.
- **BUG-002 Raw math text (v1.8.2)**. One MathJax renderer app-wide (`lib/mathjax.ts` + `MathText`); KaTeX uninstalled; service worker caches MathJax + fonts offline.
- **BUG-003 Diagram placeholders (v1.8.3)**. `diagrams` Storage bucket + policies, `lib/storage.ts` upload, numbered figure cards, Studio diagram tagging.
- **BUG-004 Generation fails on large input (v1.8.4/v1.8.5)**. Auto-split into 20k-char parts, per-part retry, typed errors (`INPUT_TOO_LARGE` , `BAD_OUTPUT` ...), async convert jobs (`202` + `poll`) so long generations survive timeouts.
- **BUG-005 Quiz leaks answers (v1.8.5)**. Submit-at-end EOQ: neutral picks, `Check score` only when all answered, verdicts/feedback/record on submit.
- **BUG-006 ME 352 not found (v1.8.6)**. `course_aliases` table + boot seed, `GET /v1/courses/search` across code/title/alias, Explore debounced search spanning both semesters with badges.
- **BUG-007 Silent stacking publishes (v1.8.7)**. Occupied week/topic returns 409 (`WEEK_OCCUPIED` / `TOPIC_OCCUPIED`); Studio offers Replace / Add as new version / Cancel. Nothing overwrites or duplicates silently.

P1 fixes (important — before the research test)

- **BUG-008 Old + new versions both listed (v1.8.7 batch)**. Student lists and the author dashboard group by topic and show the latest version only (`older` count, history in reader).
- **BUG-009 No Lecture 1/2/3 structure (v1.9.0)**. Weeks hold lectures with per-lecture topic numbering; reader switches via `?c=` with cross-lecture Back/Next; progress, versions, resume and publish are lecture-keyed (migration `0018`). Studio/NoteBuilder lecture pickers; AI prompt emits lecture groups.
Known deviation: EOQ stays one week bank after the final lecture — per-lecture banks have no storage yet (decision pending).

- **BUG-010 Partial EOQ banks (v1.9.1).** A short bank triggers one automatic repair pass requesting a fresh 10-question bank (8 MCQ + 2 FITB); repair never fails the job.
- **BUG-011 Definitions/formulas blend in (v1.9.2).** Symbol cards render through MathJax (no raw `\alpha` overlapping titles), flex-clamped, wider card spacing.
- **BUG-012 Dashboard strip (v1.9.3).** 2-column course-card grid (title, lecturer, weeks, topics done), one-tap elective removal, lecturer names batched into the catalog. **Decision pending:** BUG-012 asks for no separate Explore page — Explore stays as the add-courses path until you confirm removal (it is the only discovery route).
- **BUG-013 Slow page changes (v1.9.3).** Idle-time warmup of enrolled week-lists kills the slow-blank feel. **Note:** no discussion panel exists in the app, so that half of the report had nothing to fix.
- **BUG-014 Quiz extra click + unstyled Check (v1.9.2).** `Check score` / `Try again` styled like primary buttons (the class had zero CSS); quiz tab opens directly.

Post-list fixes

- **Explore crash (v1.9.4).** Search `useEffect` sat below the loading return → hook-count crash on load. All 14 routes audited; Explore was the sole offender.
- **Share live-links 404 (v1.11.1).** `GET /share/mine` was registered after `GET /share/:token`, so Express swallowed "mine" as a token. Route order fixed; create, resolve, revoke, WhatsApp unfurl re-verified by code path.
- **My Classes + offline-first (v1.10.0).** Recurring weekly timetable + live roster per taught course (`/classes`, migration `0019`); offline write queue (progress/quiz/resume flush on reconnect) + per-week Save offline into a version-proof cache (SW v6).
- **Google sign-in (v1.11.0).** OAuth on both auth tabs, onboarding prefill, no backend change (API is provider-agnostic).

Left + why unverified

1. **Render redeploy outstanding (blocks most verification).** Migrations `0016` (diagrams) → `0019` (class slots) plus new endpoints only land on `migrate deploy`. Until then, fixes are code-verified (`tsc` + build) but not live-verified.
2. **Google sign-in needs console work + a live test.** Supabase provider enablement and `/auth` redirect URLs cannot be done in code; after that, one real Google round-trip (new user → onboarding → dashboard) is still untested.
3. **Lecturer side never tested** (per the original list) — including the new My Classes page, which needs a teaching-account walkthrough: assign courses in admin, set slots, check the roster and the student-side class-times card.
4. **Phone pass outstanding** for student + contributor sides (layouts are mobile-first 480px by construction, but not device-tested).

5. **Progress/streak persistence retest** after the lecture + offline changes (resume now carries lecture; queue flush needs an offline→online cycle on a real device).
6. **Proposed, not built:** per-university allowed-email-domains policy (the actual gate for multi-school expansion — works with both email and Google paths), and a link-Google-to-existing-account action (same address via both methods can mint two identities).

Brand direction: why the generated kit can't hold as law

The `docs/Unify-Brand-Kit/` (posted 2 October 2026) was single-pass AI output (Claude): no user testing behind it, no ratification, and it contradicts decisions that were already tested and shipped. A generated draft is an input to a brand decision — it is not the decision. What follows is the recorded reasoning so the next reader doesn't re-litigate it from a chatbot's authority.

1. The kit contradicts the product's own retention thesis. The master PRD's core loop is Duolingo mechanics (streaks, XP, badges, tactile progression) applied to university study. The kit's headline rule — "it should never look like a game" — outlaws the very psychology the product is built on. Tested reality agrees with the PRD: residents and students shown the Duolingo-style revamp (Classic) preferred it. Playfair editorial (Story) was added as a second theme to keep the serif direction alive, not to replace a validated winner.

2. The kit contradicts tested user requests. It mandates one-column course cards; real testing (BUG-012) asked for a card grid. It bans Save PDF; the XP-gated PDF is a shipped retention incentive users understand. It bans the mascot from the dashboard; the greeting mascot is part of the daily habit loop. In each case the tested artifact wins over the generated rule — the kit's own social guideline concedes this precedent (the shipped lighter post template "is now the standard" over the earlier generated one).

3. The case for the Duolingo style (Classic, the default). Duolingo (~100M+ MAU), Quizlet and Khan Academy converge on the same findings for young mass-market learners: chunky 44px-grade touch targets (cheap Androids, touch imprecision), high-contrast CTA colour (sunlight legibility), rounded friendly type (Nunito stays legible at 12px where thin-serif Playfair strokes break on low-end screens), and celebration moments (XP bursts, streaks) that create the return habit the PRD demands. Serif-editorial is the right voice for lecturers, certificates and campaigns — not for a first-year student revising on a bus.

4. Why the theme switch stays. Two validated tastes exist and they split by role and age: students motivate on Classic; lecturers, collaborators and formal surfaces read premium on Story. Forcing one face loses one audience. Both themes persist, per-device, with usage measured — a future default follows data, not taste. Proposed instrumentation: log the active design with existing analytics events so the split is evidence, not anecdote.

5. The Unify ecosystem brand (adopted). Classic ("Play") and Story ("Editorial") are co-equal official themes, not a main and a fallback. From the kit we adopt what is genuinely good regardless of theme: plain-spoken button labels, zero hype register, visible focus states (accessibility fix queued), the social-post template structure, and Story's token set for the editorial theme. From the kit we reject: the Playfair-only rule, the gradient ban where gradients carry meaning (progress, earned states), the PDF ban, the mascot confinement, and the one-column-cards rule. Future brand proposals arrive as diffs against this section with test evidence attached — never as generated doctrine.